

CORMEN Part III Data Structures

- 1) Set: A set is a collection of elements (or objects) without duplicacy. A set that can grow and shrink over time is called dynamic.
- 2) Dictionary: A dynamic set that supports the operations of:-
 - i) inserting elements into the set
 - ii) removing elements from the set
 - iii) testing membership in a setis called a dictionary.
- 3) Each element of a set comprises of a Key (which identifies the element) and satellite data (which are carried around with the Key).
- 4) Operations on dynamic sets:

Two kinds of operations are supported by dynamic sets: querying operations (which return some information about the set) and modifying operations (which change the dynamic set)

 - (i) SEARCH(S, K): Search the set S for the Key K . Return a pointer x to an element in S such that $\text{Key}[x] = K$ or return NIL if no such element belongs to S . 1

(ii) MINIMUM(S): return a pointer to that element of set S which has the smallest key.

(iii) MAXIMUM(S): return a pointer to that element of set S which has the largest key.

(iv) INSERT(S, x): insert the element pointed to by x into the set S .

(v) DELETE(S, x): remove the element pointed to by x from the set S .

(vi) SUCCESSOR(S, x): given an element x in S , return a pointer to the next larger element in S or NIL if x is the largest element.

(vii) PREDECESSOR(S, x): given an element x in S , return a pointer to the next smaller element in S , or NIL if x is the smallest element.